



Discover Autonomous Data Intelligence

Engineering Hackathon 2026

Overview

DataGenie started in 2020 to do things the smarter way - to automate the boring and let innovation be rightfully at the center.

We are a geographically distributed team passionate about revamping the way analytics is consumed in the enterprise today. Behind the DataGenie magic is a hand-picked team of experts that has over 80 years of combined experience in data analytics. We have first-hand experience with both the 'good' & the 'not so good' parts of the current analytics landscape. While we retain & enhance the 'good', we know how exactly the 'not so good' is to be revamped.

At DataGenie, our vision is to make it effortless for businesses to realize data-driven decision-making at scale. We are agile, highly collaborative, and customer-centric in our approach to solution engineering and technology.

There's a lot that goes into building a culture and we keep it simple - work on what drives you and solve real problems alongside some of the smartest people in the business. It's real every step of the way.

We're excited to extend an invitation for you to join our Hackathon, centered around the theme ***"Lost in Translation: Teaching Machines the Language of Business"***

Put on your creative hats and get Hacking!

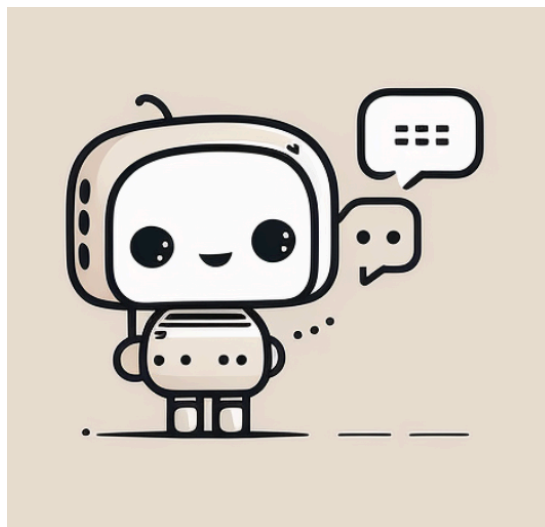


Problem Statement

Every enterprise has a warehouse full of answers and a business team full of questions, and the two rarely speak the same language. A manager asks for the “active shops in Istanbul”, but nowhere is “active” defined. There is no data dictionary and no glossary. The only clues are in the database itself: a status column, an end date that may or may not have passed, a history table where only one row is current, a status flag that quietly decides who counts as “my mobile base”.

Enter “Rosetta,” the analyst who can walk into a database it has never seen, read its tables and values, and work out what the business *means* by them. Then it answers questions and shows its working.

Can you bring “Rosetta” to life and teach a machine to learn a business from its data?



At DataGenie we build agentic analytics that let business users question enterprise data in plain language. We have learned that the LLM is the easy part. The hard part is everything around it: learning what the data means, finding the right slice of that meaning for each question, and turning it into a query that runs and returns the right numbers.

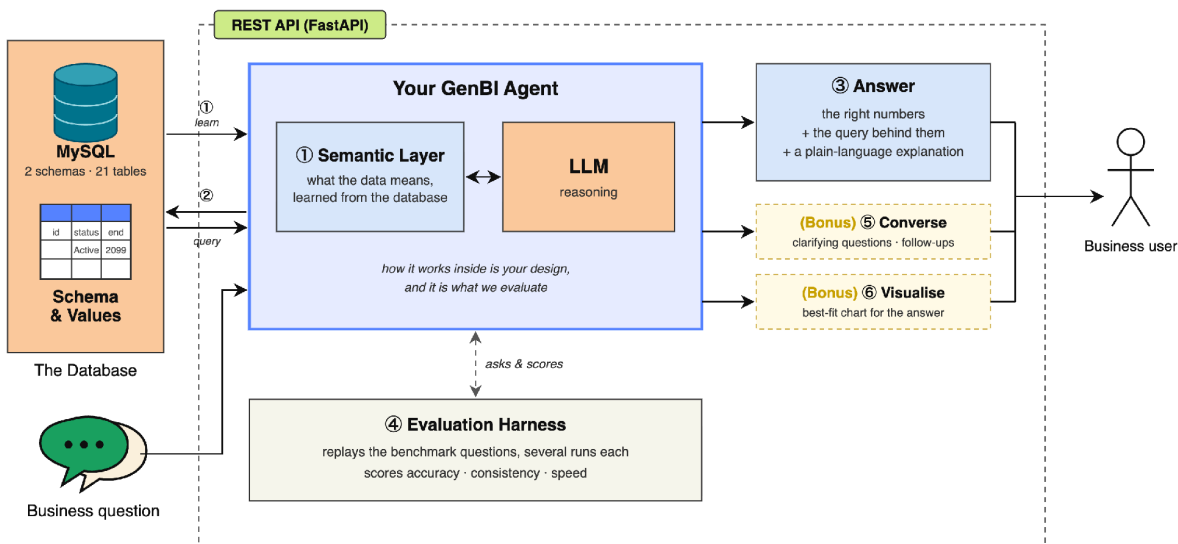
This is not a “wrap an LLM around a database” challenge. Pasting the schema into a prompt and asking for SQL will pass the easy questions and fail the hard ones, because column names do not say what the data means. The values do. We want to see a system that explores the database, learns its meaning, captures it in a semantic layer, and uses it to answer.

Goal

Build a **GenBI agent** that **learns a database from its schema and values**, captures that knowledge in a semantic layer, and answers business questions over it correctly, consistently and with evidence.



High Level Overview



The above diagram shows the big picture. How you build the inside of the agent is up to you, and it is what we evaluate:

- Checkpoint 1: Learn the Data** - Point your system at the database and let it work out what the data means: the business entities and how they relate, what the important columns and values stand for, and which records are current versus history. Capture this as a semantic layer.
Note: No data dictionary is provided. Everything should be learned from the schema and the values; if you correct anything by hand, show what changed.
- Checkpoint 2: Find the Right Data** - For every question, use what the system has learned to pick the right tables and values, and turn them into a query that runs on the database.
- Checkpoint 3: Answer with Evidence** - Return the right numbers together with the query behind them and a short plain-language explanation, including how dates such as “last month” were read.
Note: If the data cannot answer a question, say so. A wrong number is worse than no number.
- Checkpoint 4: Prove It Works** - Build an evaluation harness that runs the benchmark questions several times and reports how many answers are correct, consistent and fast.
Note: Compare the answers (the data), not the query text.
- (Bonus) Checkpoint 5: Clarify & Converse** - Ask a clarifying question when a request is genuinely ambiguous, and handle follow-ups (“now split that by city”) using the conversation so far.



6. **(Bonus) Checkpoint 6: Visualise the Answer** - From each response, suggest and generate the chart that best tells the story. You can use any common plotting library of your choice.

DB Connection Details

Use the following read-only connection details:

- **Database Type:** Oracle DB - 23ai
- **Host:** 20.102.78.61
- **Port:** 1521
- **Database name:** FREEPDB1
- **User:** VF_AGENT
- **Password:** cF9skWVHbHkMI70VMqax
- **Schemas:** VID, GENAIVF
- **Tables:** 21 in total (19 in VID, 2 in GENAIVF)

Benchmark Test Questions

Benchmark sheet: [Benchmark SQL Answer Sheet](#)

A sample from our benchmark, grouped by difficulty:

#	Level	Question	What it checks
1	Easy	List the active shops in Istanbul with their shop number, district and sales manager.	Works out what “active” means from the columns and values
2	Easy	How many phone campaigns are available for sale, by device?	Works out what “available for sale” means
3	Medium	Which customer segment had the highest number of unique customers assigned in June 2025?	Counts customers, not records
4	Medium	Compare monthly gross adds and disconnections for July and August 2026, showing the month-over-month change in each.	Finds how lines joining and leaving are recorded, then compares periods
5	Hard	As of 28 August 2026, how many subscribers in the mobile base used a physical SIM and how many used an eSIM?	Discovers who counts as the “mobile base” and reads SIM history at a point in time
6	Hard	For every disconnection in August 2026, which tariff was the subscription on at the moment of the event? Count disconnections per tariff.	Looks up the state of a line at a past moment

Note: Final evaluation uses these plus a larger **hidden set** of benchmark questions at the same levels, so hard-coded answers or per-question queries will not score.

The primary focus of this Hackathon is how your system learns the data and answers from it. A UI is not required. Having it is a plus



Rules

1. This is an **individual participation** hackathon
2. It's an **online hackathon**, you can participate from anywhere.
3. Upon receiving the problem statement you will have **3 days** to submit your implementation.
4. Use **Python** as the primary programming language for the implementation.
5. Any API written should follow REST principles. Use **FastAPI** (<https://fastapi.tiangolo.com/>) as the backend framework for the APIs.
6. You can use any freely available LLM, API-based or a self-deployed open source model. [Gemini APIs](#) have a free tier. Record which model you used.
7. Database access is **read-only**. Keep the credentials private: never share them or commit them to your repository.
8. **No hard-coding**. No per-question queries, answer lookups or question-specific prompt branches. What the system knows about the data must come from exploring it.
9. The hack must be made available on **GitHub as a private repository** giving DataGenie access to read. Provide read access to **github username: datagenie-ai**
10. To be eligible for evaluation you must have completed **Checkpoints 1, 2 and 3**.
11. The repository **must contain a README.md, a short video** explaining the implementation, and the **output of your evaluation harness**: for every question and run, the generated query, the result, the response time and the model used.
12. **Any plagiarism detected will result in immediate disqualification.**
13. **Your hack must be developed entirely during the Hackathon duration.**

Evaluation Criteria

1. **Engineering Quality** - how your system learns the data from its schema and values, how it builds and uses the semantic layer, and the quality of your evaluation harness, code and usage of git.
2. **Correct results** - the answer matches the expected data. A different but equivalent query is fully accepted; column order, row order and names are ignored.
3. **Queries that run** - every generated query runs on MySQL without manual editing.
4. **Meaning learned correctly** - terms such as “active”, “available for sale”, “mobile base” and “currently” are mapped to the right columns and values without being told.
5. **Counting at the right level** - unique customers versus records, and no double counting when data is combined.
6. **Consistency** - the same question asked three times gives the same answer.
7. **No fabrication** - no invented tables, columns or figures. When the data cannot answer, the system says so.

Note: Don't worry if you can't reach the best result or answer every benchmark question because of the limits of the LLM you use. We evaluate you more on the architecture, the evaluation harness and how you build the agentic solution than on the final score.



FAQs

1. **Do I need to have any specific qualifications to be a participant in the Hackathon?** If you love data and know how to code, you are more than welcome to participate in the Hackathon.
2. **How do I submit what I have made for the Hackathon?** Commit the source code to a private GitHub repository giving DataGenie access, along with instructions to run the application.
3. **Will you provide a database?** Yes: read-only access to a MySQL database (2 schemas, 21 tables). Connection details are shared after registration. You develop on your local system and connect remotely.
4. **Will you share a data dictionary or business glossary?** No. Learning what the data means from the schema and the values is the core of the challenge.
5. **Can I use text-to-SQL frameworks such as LangChain, LlamaIndex or Vanna?** Yes, but how the system learns the data and builds its semantic layer must be your own design. That is what we evaluate.
6. **Do I need to give a demo for the product that I have built?** Yes, a short video explaining your approach and your implementation.
7. **Is there a constraint on the frameworks I can use?** The backend must use **FastAPI**. A frontend is optional and you are free to use any framework for it.
8. **I have a question that is not listed here. Who can I contact?** You can send an email to krishnan@datagenie.ai with your queries.

*Happy Hacking! Cheers from **DataGenie Team** 😊*